SAI ABHINAV PATTOOR

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SUMMARY

Data Engineer-focused Computer Science senior graduating May 2027 (3.95 GPA). Automate production data workflows at Texas Tech, reconciling 50,000+ records weekly and saving 5+ hours per week with Python validation scripts. Project work includes scheduled AWS ETL pipelines, source-linked retrieval and MCP tools, and Node.js backends. Seeking new-grad Data Engineer and Software Engineer roles.

EDUCATION

Texas Tech University | B.S. Computer Science, Minor in Mathematics Expected May 2027

GPA 3.95/4.00 | President's Honor List: Fall 2025, Spring 2026

EXPERIENCE

Texas Tech University, Enrollment Management IT Lubbock, TX

Student Assistant, Data & Systems Engineering Feb 2026 - Present

- Automated vendor-file validation and cleanup in Python, saving **5+ hours weekly**; added regression tests for import scripts and record-resolution logic.
- Built a Python duplicate-resolution tool using deterministic matching and survivorship rules, routing ambiguous student records for human review.
- Reconcile **50,000+ records weekly** across Informatica ETL loads, checking source-to-target completeness for rejected or missing records.
- Develop and test Apex validation and synchronization logic before production to keep applicant and enrollment records consistent.

IT Systems Intern, Automation & Data Quality May - Aug 2026

- Replaced nested applicant-routing formulas with an editable lookup for **100+ programs**, so staff maintain mappings without formula changes.
- Stored reporting values with record-triggered Salesforce Flow; enforced unique record IDs to prevent duplicate migration records.

ProcureX Technologies Hyderabad, India

Technical Systems & Data Engineering Intern Jun - Aug 2025

- Validated **40+ procurement data flows** across **8 components** using Python and Excel; caught and documented **15 part-ID and unit-of-measure mismatches** before integration.

PROJECTS

Austin Real Estate Analyzer | Python, Prefect, SQLite, AWS Dec 2025 - Nov 2026

- Built a weekly Python/Prefect ETL pipeline; scheduled AWS CodeBuild through EventBridge and defined infrastructure in CloudFormation.
- Enabled Streamlit dashboard recovery from private S3 via CloudFront without app AWS credentials; added SHA-256/SQLite integrity checks and **12 pipeline/reader tests**.

ZipRAG | Python, SQLite FTS5, MCP Nov 2026

- Indexed **18 Austin articles** as **109 passages** using SQLite FTS5 and TF-IDF; returned excerpts with source URLs and revision metadata.
- Exposed document search and dated ZIP-score lookup through read-only MCP tools; added **16 tests** for tool calls, validation, source integrity, and client/server handshakes.

The Prospector | Node.js, Express.js, MongoDB Sep 2025

- Co-developed a **HackWesTX runner-up** search app; built an Express.js backend that combines external API results into a common response format.
- Designed a MongoDB saved-search schema; implemented URL-based deduplication, per-source error handling, and request timeouts.

LEADERSHIP

Marketing Lead | Google Developer Groups at Texas Tech Jan - Dec 2025

- Led chapter marketing and helped host technical workshops for student developers.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript
Data & Backend: pandas, Informatica, Prefect, Node.js/Express, REST APIs, SQLite, MongoDB
Cloud & Tools: AWS (S3, CodeBuild, EventBridge, CloudFront, CloudFormation), Git/GitHub, Docker, Linux
Frameworks & Platforms: React, Streamlit, scikit-learn, MCP, Salesforce (Apex, Flow, SOQL)